21/08/2024

Code-B_Phase-2

Aakash
Medical|IIT-JEE|Foundations

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MM : 720 NCERT Booster Test Series for NEET (RM)-2024-25_T01B **Time : 200 Min.**

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy & Power, System of Particles & Rotational Motion
Chemsitry: Some Basic Concepts of Chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Botany: Cell : The Unit of Life, Cell Cycle & Cell Division, The Living World, Biological Classification
Zoology: Structural Organisation in Animals–Animal Tissues, Biomolecules, Breathing & Exchange of Gases, Body Fluids & Circulation

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt **all 35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

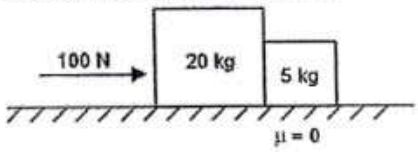
PHYSICS

SECTION-A

1. The area under the curve of acceleration time graph represents:
   (1) Velocity
   (2) Speed
   (3) Change in position
   (4) Change in velocity

2. Position of a particle varies as $x = 2t + 3t^2$ (where x is in m and t is in s) the average velocity of the particle in time interval 1 s to 2 s is
   (1) 11 m/s
   (2) 9 m/s
   (3) 13 m/s
   (4) 6.5 m/s

3. Find contact force between blocks.



   (1) 20 N
   (2) 40 N
   (3) 60 N
   (4) 50 N

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4. A : The moment of inertia is not a vector quantity.
R : Moment of Inertia of a rigid body depends on mass distribution and position of axis of rotation.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

5. The instantaneous angular position of a point on rotating wheel is given as $\theta = 4t^3 - 12t^2$. The torque on the wheel is zero at

(1) 1 s

(2) 1.5 s

(3) 2 s

(4) 2.5 s

6. A body is moving with velocity 30 m/s towards east. After 10 seconds its velocity becomes 40 m/s towards north. The average acceleration of the body is

(1) 5 m/s^2

(2) 1 m/s^2

(3) 7 m/s^2

(4) $\sqrt{7}$ m/s^2

7. Two balls A and B are moving with velocity $\left(\hat{i}+\hat{j}\right)$ m/s and $\left(4\hat{i}-3\hat{j}\right)$ m/s respectively. Then the magnitude of relative velocity of ball B w.r.t. ball A is

(1) 3 m/s

(2) 4 m/s

(3) $\sqrt{41}$ m/s

(4) 5 m/s

8. If error in measurement of force exerted on a circular plate of negligible thickness is 4% and in the measurement of pressure is 1%, then percentage error in calculated value of radius of circular plate will be

(1) 3%

(2) 1.5%

(3) 2.5%

(4) 5%

9. If force (F), velocity (V) and time (T) are taken as fundamental units, then dimension of length are

(1) $[F^1V^1T^1]$

(2) $[F^0V^1T^1]$

(3) $[F^{-1}V^{-1}T^{-1}]$

(4) $[F^0V^{-1}T^{-1}]$

10. If $a = 5.0 \pm 0.1$ and $b = 5.0 \pm 0.2$ then $2b - a$ is equal to

(1) 5.0 ± 0.3

(2) 5.0 ± 0.5

(3) 15.0 ± 0.5

(4) 15.0 ± 0.3

11. If in a vernier callipers 5 VSD coincides with 4 MSD, then least count of vernier calliper is (given 1 MSD = 1 mm)
(VSD → vernier scale)
(MSD → main scale division)

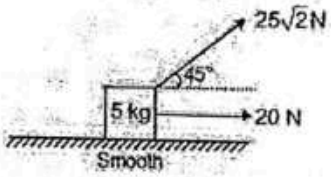
(1) 0.1 mm

(2) 0.4 mm

(3) 0.3 mm

(4) 0.2 mm

12. The acceleration of the given block is



(1) 3 m/s^2

(2) 9 m/s^2

(3) 4 m/s^2

(4) 5 m/s^2

13. A ball is dropped from a height of 100 m. Due to collision with the floor, it loses 20% of its kinetic energy. After collision, ball will rise to a height of

(1) 80 m

(2) 20 m

(3) 40 m

(4) 50 m

14. Work done by conservative forces acting on body is always equal to

(1) Negative of change in potential energy

(2) Magnitude of change in potential energy

(3) Change in kinetic energy

(4) Change in mechanical energy

15. Moment of inertia of a spinning object drops to $\left(\frac{1}{3}\right)^{rd}$ of its initial value. The ratio of new rotational kinetic energy to initial rotational kinetic energy is (Assume net external torque about the axis of rotation is zero)

(1) 1 : 3

(2) 3 : 1

(3) 3 : 2

(4) 2 : 3

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16. A body is projected at 60° from horizontal with ground. It covers a horizontal distance of 100 m. If the same body is projected at 60° with vertical with same velocity, the new range is

(1) 50 m

(2) 100 m

(3) 200 m

(4) 150 m

17. A disc rotates about its axis from rest, acquires the angular speed 40 rev/s in 4 seconds with constant angular acceleration. The angle rotated by it (in radians) during $t = 2$ s to $t = 4$ s is

(1) 30π

(2) 60π

(3) 120π

(4) 180π

18. During somersault, a swimmer bends his body to

(1) Increase moment of inertia

(2) Decrease moment of inertia

(3) Increase angular momentum

(4) Decrease angular speed

19. A body of mass 2 kg is under a constant force which causes a displacement s (in metres) as $s = \frac{1}{2}t^2$, where t is in second. The work done by the force in 2 s is

(1) 2 J

(2) 4 J

(3) 1 J

(4) 6 J

20. If a force $\vec{F} = (\hat{i} + 2\hat{j} - 3\hat{k})$ N is applied at a point whose position vector is $\vec{r} = (\hat{i} - 2\hat{j} + \hat{k})$ m, then unit vector along torque vector about the origin will be

(1) $\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$

(2) $\frac{4}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

(3) $\frac{\hat{i}-\hat{j}+\hat{k}}{\sqrt{3}}$

(4) $\frac{4}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$

21. A 500 kg car takes a round turn of radius 50 m with a velocity of 36 kmph, the centripetal force is

(1) 250 N

(2) 1000 N

(3) 750 N

(4) 1200 N

22. A boat is moving with a velocity of $\left(2\hat{i} + 3\hat{j}\right)$ m/s with respect to ground. The water in the river is moving with velocity of $\left(-2\hat{i} - 2\hat{j}\right)$ m/s with respect to ground. The relative velocity of boat with respect to water is

(1) $\left(4\hat{i} + 5\hat{j}\right)$ m/s

(2) $4\hat{i}$ m/s

(3) $5\hat{j}$ m/s

(4) $\left(-4\hat{i} - 5\hat{j}\right)$ m/s

23. A car of mass m is moving on a level circular track of radius R. If μ_s represents the static friction between the road and tyres of the car, the maximum speed of the car in circular motion is

(1) $\sqrt{mRg}/\mu_s$

(2) $\sqrt{\mu_s Rg}$

(3) $\sqrt{\mu_s mRg}$

(4) $\sqrt{Rg/\mu_s}$

24. A particle of mass 2 kg is projected at an angle 60^0 with horizontal with a speed 50 m/sec, find radius of curvature at highest point of trajectory ($g = 10$ m/s^2).

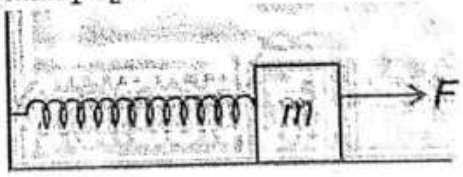
(1) 62.5 m

(2) 6.25 m

(3) 700 m

(4) 90 m

25. A constant force F acts on a block connected to spring of force constant K as shown. The extension obtained in equilibrium is x_1. If the spring constant is quadrupled, then extension in equilibrium with same force F is x_2. Then $x_1 : x_2$ is



(1) 1 : 2

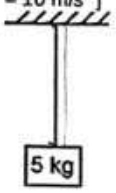
(2) 1 : 4

(3) 1 : 3

(4) 4 : 1

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26. In the shown figure, mass of the rope is 10 kg and of the block is 5 kg. The tension in the rope at topmost point is [$g = 10\ m/s^2$]



(1) 150 N

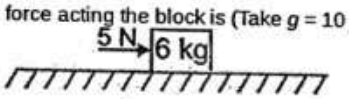
(2) 50 N

(3) 100 N

(4) 75 N

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27. A block of mass 6 kg lying on a rough horizontal surface with coefficient of static friction ($\mu_s = 0.2$) is being acted upon by a horizontal force of 5 N as shown. The frictional force acting the block is (Take $g = 10\ ms^{-2}$)



(1) 5 N

(2) 7 N

(3) 9 N

(4) 12 N

28. A particle is in non-uniform circular motion. Which of the following are perpendicular to the velocity?

(1) Net acceleration

(2) Radius vector

(3) Tangential acceleration

(4) All of these

29. Potential energy of a particle (in J) at position x is given by $U = x^2 - 10x$. Force on the particle at $x = 0$ is (x is in m and F is in N)

(1) 10 N

(2) 0 N

(3) 2.5 N

(4) 7 N

30. The work done by kinetic frictional force may be

(1) Negative

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(2) Positive

(3) Zero

(4) All of these

31. Two springs have their force constant as k_1 and k_2 ($k_1 > k_2$). When they are stretched by the same force

(1) No work is done by this force in case of both the springs

(2) Equal work is done by this force in case of both the springs

(3) More work is done by this force in case of second spring

(4) More work is done by this force in case of first spring

32. The dimensional formula for electric potential is

(1) $[M L^2 T^{-2} A]$

(2) $[M L^2 T^{-3} A]$

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(3) $[M L^2 T^{-3} A^{-1}]$

(4) $[M L^2 T^{-2} A^{-1}]$

33. The dimensions of impulse is same as that of

(1) Linear Momentum

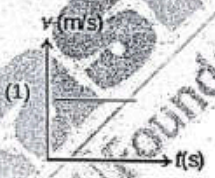
(2) Angular Momentum

(3) Plane Angle

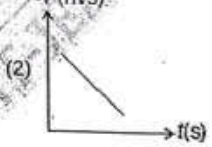
(4) Energy

34. The speed-time graph representing a uniform rectilinear motion is

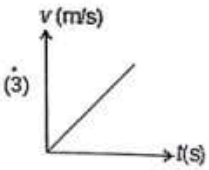
(1)



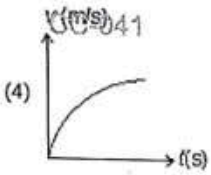
(2)



(3)



(4)



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35. A body travels half of its total path in the last second of its fall from rest then time of fall is ($g = 9.8\ m/s^2$)

(1) 3.4 s

(2) 3.7 s

(3) 4.2 s

(4) 4.5 s

SECTION-B

36. **Assertion (A):** Two physical quantities having same dimension always have same unit.
Reason (R): A dimensionless quantity is always unitless.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

37. A wheel is rotating with an angular acceleration 2 rad/s^2. If initial angular velocity of wheel is 2 rad/s, then angular velocity of wheel at $t = 4$ s is

(1) 2 rad/s

(2) 6 rad/s

(3) 8 rad/s

(4) 10 rad/s

38. The velocity of a bullet is reduced from 400 ms^{-1} to 100 ms^{-1} while travelling through material of thickness of 1 m. The average retardation will be

(1) 7.5×10^4 m/s^2

(2) 8.5×10^3 m/s^2

(3) 7.5×10^2 m/s^2

(4) 8.5×10^2 m/s^2

39. Power delivered to a particle varies with the time as $P = (2t + 3)$ W, where t is time in second. The change in kinetic energy between time $t = 0$ s to $t = 2$ s will be

(1) 20 J

(2) 5 J

(3) 10 J

(4) 15 J

40. A coin placed on a rotating turntable just slips if it is placed at a distance of 4 cm from the centre. If the angular velocity of turntable is doubled, it will just slip at a distance of

(1) 1 cm

(2) 2 cm

(3) 3 cm

(4) 4 cm

41. A particle moves from a point $(-2\hat{i}+5\hat{j})$ to $(4\hat{j}+3\hat{k})$ when a force of $(4\hat{i}+3\hat{j})$ N is applied. How much work has been done by the force?

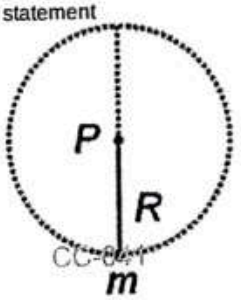
(1) 8 J

(2) 11 J

(3) 5 J

(4) 2 J

42. A bob is whirled in a vertical circular motion of radius R using a string as shown in figure. Choose the correct statement



(1) Net work done by gravity is always zero at any instant

(2) Net work done by tension is always zero at any instant

(3) Net work done by both gravity and tension is always zero at any instant

(4) Net work done by both gravity and tension is always non zero at any instant

43. A ball of mass 2 kg is moving towards batsman with a speed 30 ms^{-1}. The batsman strikes the ball and deflected it through an angle of 60° without change in speed. The impulse imparted to the ball is

(1) $20\sqrt{3}$ Ns

(2) $30\sqrt{3}$ Ns

(3) $60\sqrt{3}$ Ns

(4) 40 Ns

44. A particle is projected with a velocity u making an angle θ with the horizontal. The speed of particle when particle makes angle ϕ with horizontal is

(1) $u \cos\theta \cos\phi$

(2) $u \cos\theta \sec\phi$

(3) $u \cos\theta \operatorname{cosec}\phi$

(4) $u \cos\theta \sin\phi$

45. A projectile is projected with initial velocity $\left(6\hat{i}+8\hat{j}\right)$ m/s. Then its horizontal range is ($g = 10$ m/s^2)

(1) 9.6 m

(2) 4.9 m

(3) 19.2 m

(4) 4.8 m

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46. A flywheel is rotating about a fixed axis and has kinetic energy of 450 J, when its angular speed was 30 rad s^{-1}. The moment of inertia of wheel about the fixed axis is

(1) 1 kg m^2

(2) 2 kg m^2

(3) 0.5 kg m^2

(4) 4 kg m^2

47. A ball is thrown vertically upward with speed 20 m/s. The ball is above 15 m height for duration (g = 10 m/s^2)

(1) 1 s

(2) 2 s

(3) 1.5 s

(4) 3 s

48. The distance travelled by a particle is directly proportional to $\sqrt{t}$, where t is time elapsed. Which of the following is correct regarding the acceleration a of the particle?

(1) $a \propto t^{\frac{5}{2}}$

(2) $a \propto t^{\frac{3}{2}}$

(3) $a \propto t^{\frac{-3}{2}}$

(4) $a \propto t^{\frac{-5}{2}}$

49. An object is projected horizontally with the speed of 10 m/s from the top of a building at height 20 m. The distance between bottom of the building and point where particle strikes on ground will be (g = 10 m/s^2)

(1) 10 m

(2) 40 m

(3) 20 m

(4) 30 m

50. At a place rain is falling vertically down with a velocity of 4 m/s. A person running on a horizontal plane feels rain is falling at 5 m/s. The velocity of the person is

(1) 4 m/s

(2) 3 m/s

(3) 5 m/s

(4) 2.8 m/s

CHEMISTRY

SECTION-A

51. Identify the example of pure substance among the following

(1) Milk

(2) Air

(3) Sugar solution

(4) Carbon dioxide

52. Identify the incorrect match.

	Multiple	Prefix
(1)	10^{-18}	atto
(2)	10^{-15}	pico
(3)	10^{-21}	zepto
(4)	10^{-9}	nano

(1) (1)

(2) (2)

(3) (3)

(4) (4)

53. Correct scientific notation is

(1) 0.000473

(2) 0.473×10^{-3}

(3) 4.73×10^{-4}

(4) 473×10^{-6}

54. An element X has the following isotopic composition. ^{100}X: 80%, ^{102}X : 20%.
The weighted average atomic mass of the naturally occurring element X is closest to

(1) 101.1 u

(2) 100.4 u

(3) 100.8 u

(4) 101.8 u

55. If the mass ratio of helium and methane is 1 : 8 then the ratio of the moles of two gases will be

(1) 1 : 8

(2) 1 : 4

(3) 1 : 1

(4) 1 : 2

56. A molecule contains 8% sulphur by mass. Minimum molecular mass of the molecule will be

(1) 200 u

(2) 400 u

(3) 600 u

(4) 800 u

57. 6.5 g of an impure sample of limestone liberates 2.2 g of CO_2 on strong heating. The percentage purity of $CaCO_3$ in the sample is

(1) 85.2%

(2) 76.9%

(3) 72.5%

(4) 92.5%

58. A sample of aluminium sulphate, $Al_2(SO_4)_3$ contains 2 moles of sulphur atoms, the number of moles of oxygen atoms in the sample is

(1) 2

(2) 3

(3) 6

(4) 8

59. Two elements A(Atomic mass = 60 u) and B(Atomic mass = 20 u) combine to give a compound having 75% of A by mass. The empirical formula of compound is

(1) A_2B

(2) AB

(3) AB_2

(4) A_2B_3

60. Which of the following is not a subatomic particle?

(1) Electron

(2) Neutron

(3) Proton

(4) Alpha particle

61. Incorrect statement about cathode rays is

(1) These are negatively charged particles

(2) They originate from cathode and move towards anode

(3) Characteristics of cathode rays particles depend upon the nature of gas present in the tube

(4) In absence of electric or magnetic field, these rays travel in straight lines

62. Which group represents isoelectronic species?

(1) Cl^-, Na^+, Ne, Mg^{2+}

(2) Cl^-, Ne, Al^{3+}, Ca^{2+}

(3) F^-, Cl^-, Ne, Ar

(4) Na^+, F^-, Ne, Al^{3+}

63. If the wavelength of electromagnetic radiation is 100 nm then the energy of the quanta will be ($h = 6.625 \times 10^{-34}$ J s; $c = 3 \times 10^8$ ms^{-1})

(1) 6.25×10^{-18} J

(2) 3.75×10^{-19} J

(3) 6.625×10^{-17} J

(4) 1.99×10^{-18} J

64. Orbital angular momentum of electron in *p* orbital is equal to

(1) 0

(2) $\sqrt{6}\hbar$

(3) $\sqrt{2}\hbar$

(4) $2\sqrt{3}\hbar$

65. Given below are two statements.

Statement I : The radius of 2nd orbit of He^+ ion is 1.058 Å.

Statement II : Splitting of spectral line in electrical field is called stark effect.

Choose the correct option from the following:

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statements are incorrect

(4) Both statements are correct

66. If a particle of mass 0.5 g is moving with a velocity of 200 m/s then the wavelength associated with the particle is ($h = 6.625 \times 10^{-34}$ Js)

(1) 6.625×10^{-32} m

(2) 6.625×10^{-33} m

(3) 6.625×10^{-34} m

(4) 6.625×10^{-36} m

67. Consider the following statements

(a) Variation of heat capacity of solids can be explained by particle nature of electromagnetic radiation

(b) Diffraction of light can be explained by wave nature of electromagnetic radiation

(c) Work function of lithium is lesser than sodium

The correct statement(s) is/are

(1) (a) only

(2) (a) and (b) only

(3) (b) and (c) only

(4) (a), (b) and (c)

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68. Out of the following set of quantum numbers, choose the one which does not exist

(1) $n = 1, l = 0, m = 0, s = -\frac{1}{2}$

(2) $n = 4, l = 0, m = 0, s = +\frac{1}{2}$

(3) $n = 3, l = 3, m = 0, s = +\frac{1}{2}$

(4) $n = 3, l = 0, m = 0, s = -\frac{1}{2}$

69. Out of the following oxides choose the one which reacts with both acid and base?

(1) Na_2O

(2) Al_2O_3

(3) SiO_2

(4) MgO

70. According to modern periodic table the element with atomic number 31 belongs to

(1) Group = 14 and period = 4

(2) Group = 13 and period = 3

(3) Group = 15 and period = 3

(4) Group = 13 and period = 4

71. The IUPAC official name of the element with atomic number 106 is

(1) Mendelevium

(2) Dubnium

(3) Seaborgium

(4) Bohrium

72. The electronic configuration of an element (X) is $1s^2, 2s^2 2p^6, 3s^1$. The atomic number of the element which is just below X in the periodic table is

(1) 19

(2) 20

(3) 18

(4) 12

73. Match the column and choose the correct option.

Column I (Electronic configuration)	Column II (Block)
a. $2s^1$	(i) *d*-block
b. $3s^2 3p^4$	(ii) *s*-block
c. $3s^2 3p^6 3d^5 4s^2$	(iii) *f*-block
d. $4f^7 5d^1 6s^2$	(iv) *p*-block

(1) a(ii), b(iv), c(i), d(iii)

(2) a(iv), b(ii), c(i), d(iii)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(ii), b(iv), c(iii), d(i)

74. Correct decreasing order of ionic radii is

(1) $Na^+ > Mg^{2+} > F^- > O^{2-}$

(2) $O^{2-} > F^- > Na^+ > Mg^{2+}$

(3) $Mg^{2+} > Na^+ > F^- > O^{2-}$

(4) $O^{2-} > F^- > Mg^{2+} > Na^+$

75. For the second period elements the correct increasing order of first ionisation enthalpy is:

(1) Li < Be < B < C < N < O < F < Ne

(2) Li < B < Be < C < O < N < F < Ne

(3) Li < B < Be < C < N < O < F < Ne

(4) Li < Be < B < C < O < N < F < Ne

76. The valence electrons in the atom of element A is 3 and in the atom of element B is 7. The most probable compound formed from A and B is

(1) AB_3

(2) A_3B

(3) A_2B_3

(4) A_3B_2

77. Among the following, choose an incorrect statement.

(1) XeF_4 molecule is square planar in shape

(2) Ethylene is a planar molecule

(3) HF forms intermolecular hydrogen bonding

(4) O_2^+, O_2^{2-}, O_2^- all are paramagnetic in nature

78. Which of the following is an exception to the octet rule?

(1) PCl_3

(2) SCl_2

(3) SF_6

(4) CO_2

79. Which among the following has highest lattice enthalpy?

(1) LiCl

(2) NaCl

(3) LiI

(4) NaI

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80. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A): Canonical structures of a molecule are not the correct representation of the real molecule.
Reason (R): The energy of resonance hybrid of a molecule is higher than that of any of the canonical structures.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Assertion & Reason are true and reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but reason is not the correct explanation of the assertion

(3) Assertion is true but Reason is false

(4) Assertion and Reason are false

81. Which among the following is bent shape molecule?

(1) H_2O

(2) NH_3

(3) BF_3

(4) $BeCl_2$

82. Which of the following has pπ - dπ bonding?

(1) H_2CO_3

(2) H_3PO_3

(3) HNO_3

(4) H_3BO_3

83. Find the correct match.

	Column I		Column II
a.	BCl_3	(i)	sp^2
b.	$SiCl_4$	(ii)	sp^3d
c.	SF_4	(iii)	sp^3d^2
d.	BrF_5	(iv)	sp^3

(1) a(i), b(ii), c(iii), d(iv)

(2) a(i), b(iv), c(ii), d(iii)

(3) a(i), b(ii), c(iv), d(iii)

(4) a(iv), b(i), c(ii), d(iii)

84. Hydrogen bonding is absent in

(1) H_2O

(2) NH_3

(3) HCl

(4) CH_3OH

85. The correct order of decreasing negative electron gain enthalpy is

(1) F > Cl > Br > I

(2) Cl > F > Br > I

(3) Cl > F > I > Br

(4) Br > Cl > F > I

SECTION-B

86. When 100 mL of 17% (w/v) solution of $AgNO_3$ is mixed with 100 mL of 5.85% (w/v) NaCl solution, the mass of precipitate formed will be (Ag = 108, N = 14, Na = 23, Cl = 35.5)

(1) 14.35 g

(2) 27.5 g

(3) 8.35 g

(4) 11.35 g

87. If 160 g of NaOH is present in 500 mL of the aqueous solution, then the molarity of the solution will be

(1) 2 M

(2) 8 M

(3) 6 M

(4) 5 M

88. 5.6 g of nitrogen and 2 g of hydrogen were heated in a closed steel vessel. Amount of ammonia produced in this reaction can be

(1) 6.8 g

(2) 8.5 g

(3) 3.4 g

(4) 4.25 g

89. The number of atoms in 5.6 litre CH_4 at NTP is

(1) 7.52×10^{23}

(2) 6.01×10^{23}

(3) 2.30×10^{24}

(4) 6.5×10^{24}

90. Isobars are the atoms with

(1) Different atomic number and mass number

(2) Same atomic number but different mass number

(3) Same atomic number and mass number

(4) Same mass number but different atomic number

91. Correct order of decreasing wavelength is

(1) UV > Visible > IR

(2) UV > IR > Visible

(3) Visible > IR > UV

(4) IR > Visible > UV

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92. Maximum number of electrons associated with $\ell = 1$, m = 0 in magnesium atom is

(1) 1

(2) 2

(3) 6

(4) 8

93. General electronic configuration of lanthanoids is

(1) $(n-2)f^{1-14}(n-1)d^{0-2}ns^1$

(2) $(n-2)f^{1-13}(n-1)d^{0-2}ns^2$

(3) $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$

(4) $(n-2)f^{1-12}(n-1)d^{0-4}ns^2$

94. Correct covalent bond length order in C – C, O – H, C – O & C – H is

(1) C – C > C – O > C – H > O - H

(2) C – C > C – H > O – H > C – O

(3) C – O > C – C > O – H > C – H

(4) C – H > O - H > C – C > C – O

95. Aluminium shows diagonal relationship with

(1) Li

(2) B

(3) Mg

(4) Be

96. Element of highest negative electron gain enthalpy among the following is

(1) He

(2) Ne

(3) Na

(4) S

97. C – O bond order in carbonate ion is

(1) 1.25

(2) 1.33

(3) 1.50

(4) 1.75

98. The hybridisation of P in PCl_5 is

(1) sp^3

(2) dsp^2

(3) sp^3d

(4) dsp^3

99. In which of the given molecules only π -bonds are present?

(1) O_2

(2) C_2

(3) N_2

(4) H_2

100. Most electropositive element among the following is

(1) Na

(2) Mg

(3) Al

(4) Be

BOTANY

SECTION-A

101. Endomembrane system of cell includes all, except

(1) Golgi bodies

(2) ER

(3) Mitochondria

(4) Lysosomes

102. The plasmid DNA

(1) Is found in all eukaryotes

(2) Confers certain unique phenotypic characters to bacteria

(3) Is the genetic material inside the genomic DNA of prokaryotes

(4) Is larger than genomic DNA

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103. Select the **mismatched** pair.

(1) Vacuole – Tonoplast

(2) Inclusion body – Found only in eukaryotic cells

(3) Chromatophores – Blue-green algae

(4) Mitochondria – Sausage-shaped

104. Plastid which is specific for storage of oils and fats is

(1) Amyloplast

(2) Aleuroplast

(3) Elaioplast

(4) Chromoplast

105. Match the following columns and select the **correct** option.

	Column I		Column II
a.	Anton Von Leeuwenhoek	(i)	Fluid mosaic model of plasma membrane
b.	Rudolf Virchow	(ii)	First described nucleus as a cell organelle
c.	Robert Brown	(iii)	First saw and described a live cell
d.	Singer and Nicolson	(iv)	*Omnis cellula-e cellula*

(1) a(iii), b(iv), c(ii), d(i)

(2) a(ii), b(iv), c(iii), d(i)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(iv), b(ii), c(i), d(iii)

106. Which of the following statement is **incorrect**?

(1) Inner membrane of mitochondria forms a number of foldings called cristae

(2) Middle lamella of plants is composed of chitin

(3) Plasmodesmata connect the neighbouring plant cells

(4) Mitochondria are the sites of aerobic respiration

107. Cytoskeleton present in the cytoplasm is an elaborate network of filamentous structures made up of

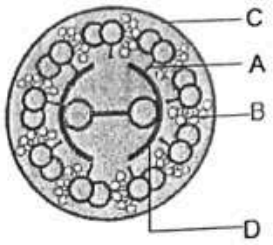
(1) Branched lipids

(2) Unbranched straight chain lipids

(3) Proteins

(4) Carbohydrates

108. Observe the diagram given below and choose the **correct** option.



(1) A – Radial spoke, B – Interdoublet bridge, C – Central sheath, D – Plasma membrane

(2) A – Central spoke, B – Interdoublet bridge, C – Central Microtubule, D – Plasma membrane

(3) A – Radial spoke, B – Interdoublet bridge, C – Central sheath, D – Central Microtubule

(4) A – Radial spoke, B – Interdoublet bridge, C – Plasma membrane, D – Central sheath

109. State true **(T)** or false **(F)** for given statements and select the **correct** option.

(a) Nucleoli are larger and numerous in the cells actively involved in protein synthesis.

(b) Outer membrane of nucleus is often connected to Golgi apparatus.

(c) Chromatin fibre could be stained with basic dyes.

	(a)	(b)	(c)
(1)	T	T	T
(2)	F	T	T
(3)	T	F	T
(4)	F	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

110. a. Maintenance of the shape of cell

b. Mechanical support

c. Motility

The above given functions are performed by

(1) Cilia and Flagella

(2) Microbodies

(3) Cytoskeleton

(4) ER

111. Select the **incorrect** option about G_0 phase.

(1) It is also called quiescent stage

(2) Cells in this stage are metabolically inactive

(3) Cells do not proliferate unless called on to do so

(4) Cells exit G_1 phase to enter this phase

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112. During meiosis, synaptonemal complex formation takes place in which of the following stages?

(1) Leptotene

(2) Zygotene

(3) Diplotene

(4) Diakinesis

113. _____ marks the end of the cell division.
Select the **correct** option to complete the statement.

(1) Formation of metaphasic plate

(2) Division of nucleus into two daughter nuclei

(3) Reappearance of Golgi bodies in the cells

(4) Division of cytoplasm into two daughter cells

114. Duration of one cell cycle for yeast cell is about

(1) 90 hours

(2) 24 minutes

(3) 90 minutes

(4) 24 hours

115. What will be the amount of DNA in each meiosis II product if the mother cell has 40 pg of DNA in G_2 phase?

(1) 10 pg

(2) 20 pg

(3) 40 pg

(4) 80 pg

116. A tetrad formed during meiosis is composed of

(1) 4 pairs of homologous chromosomes with 4 chromatids

(2) 2 pairs of homologous chromosomes with 2 chromatids

(3) A pair of homologous chromosomes with 4 chromatids

(4) 2 pairs of non-homologous chromosomes

117. Meiosis involves

(1) Two sequential cycles of DNA replication

(2) Only one cycle of nuclear division

(3) One cycle of DNA replication

(4) Two cycles of nuclear division and only one cycle of cytoplasm division

118. Select the **correct** statement.

(1) In all organisms gametes are produced only by meiosis

(2) In plants meiosis takes place in diploid mother cell

(3) Mitosis cannot be a method for gamete formation

(4) Meiosis can occur in haploid cells

119. There are five different stages of prophase-I i.e., Leptotene (L), Pachytene (P), Diplotene (D), Zygotene (Z) and Diakinesis (D_i).
Arrange the stages in **correct** sequence.

(1) $D \rightarrow P \rightarrow L \rightarrow D_i \rightarrow Z$

(2) $L \rightarrow Z \rightarrow P \rightarrow D \rightarrow D_i$

(3) $Z \rightarrow L \rightarrow D_i \rightarrow P \rightarrow D$

(4) $P \rightarrow L \rightarrow Z \rightarrow D_i \rightarrow D$

120. Growth and reproduction are synonymous for

(1) Higher eukaryotes

(2) Unicellular organisms

(3) Multicellular eukaryotes

(4) Higher plants and animals

121. Select the **incorrect** match w.r.t. taxonomic categories of wheat.

(1) Family – Poaceae

(2) Order – Monocotyledonae

(3) Division – Angiospermae

(4) Genus – *Triticum*

122. In which among the following taxonomic categories, organisms share less number of common characters than that of order?

(1) Family

(2) Species

(3) Class

(4) Genus

123. ICBN stands for

(1) International Code for Bacterial Nomenclature

(2) Indian Code for Botanical Nomenclature

(3) International Criteria for Biological Nomenclature

(4) International Code for Botanical Nomenclature

124. **Statement-A:** Both the words "Taxonomy" and "Systematics" have been derived from Greek words.
Statement-B: Systematics is the science which deals with diversity of organisms and evolutionary relationships among them.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements are correct

(4) Both the statements are incorrect

125. The number of species that are known and described range between

(1) 1.7 – 1.8 billion

(2) 1.7 – 1.8 million

(3) 7 – 8 million

(4) 17 – 18 billion

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126. Regarding the features of *Mycoplasma*, choose the option which is not true for it.

(1) It is a unicellular organism

(2) It cannot survive without oxygen

(3) It is the smallest living cell

(4) It can be pathogenic in plants

127. Archaebacteria found in saline conditions are

(1) Thermoacidophiles

(2) Halophiles

(3) Methanogens

(4) Cyanobacteria

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128. Which of the following is not the basis of classification of fungi?

(1) Morphology of the mycelium

(2) Mode of spore formation

(3) Mode of nutrition

(4) Fruiting bodies

129. *Nostoc* lacks

(1) Heterocysts

(2) Mucilagenous sheath

(3) Photosynthetic pigments

(4) Heterotrophic mode of nutrition

130. Lichens are symbiotic association of

(1) Algae and fungi

(2) Algae and bacteria

(3) Plant and fungi

(4) Bacterium with roots of plants

131. The three-domain system of classification divides the

(1) Kingdom Monera into three domains

(2) Eukaryotes into two domains

(3) Kingdom Monera into two domains

(4) Kingdom fungi into two domains

132. Which of the following statements is **incorrect** regarding Chrysophytes?

(1) They include both diatoms and desmids

(2) They are microscopic and float passively in water current

(3) They belong to kingdom Protista

(4) Most of them are heterotrophs

133. Identify the **correct** statements w.r.t. infectious agent discovered by T.O. Diener.

A. It is larger than virus
B. It is low molecular weight RNA
C. It causes potato spindle tuber disease
D. It has a protein coat called capsid

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(1) A and D

(2) B and C

(3) B and D

(4) A and C

134. Choose the **mis-matched** pair

	Protozoans	–	Organism
A.	Ciliated	–	*Paramoecium*
B.	Sporozoan	–	*Entamoeba*
C.	Flagellated	–	*Trypanosoma*
D.	Amoeboid	–	*Amoeba*

(1) A

(2) B

(3) C

(4) D

135. Select the **correct** statement w.r.t. the members of Phycomycetes.

(1) Hyphal wall contains chitin only

(2) All of them are found in terrestrial habitat

(3) The mycelium is multinucleated and aseptate

(4) Zoospores found in their members, are non-motile

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SECTION-B

136. Choose the **correct** hierarchical arrangement in ascending order.

(1) Genus → Family → Class → Order

(2) Species → Genus → Order → Family

(3) Order → Class → Phylum → Kingdom

(4) Species → Family → Division → Class

137. Read the following statements and choose the **correct** option.
Assertion : During zygotene stage, chromosomes start pairing together and this process is called terminalisation.
Reason : Anaphase-I begins with simultaneous splitting of centromeres of each chromosome.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

138. Which of the following components provide the fluid nature to plasma membrane?

(1) Proteins

(2) Cholesterols

(3) Carbohydrates

(4) Phospholipids

139. Which structure prevents the bacteria from bursting and collapsing?

(1) Slime layer

(2) Capsule

(3) Plasma membrane

(4) Cell wall

140. No virus contains both

(1) RNA and protein coat

(2) DNA and protein coat

(3) RNA and DNA

(4) Capsid and double stranded DNA

141. The most dramatic period of the cell cycle is

(1) S-phase

(2) M-phase

(3) G_2-phase

(4) G_1-phase

142. Main criteria for classification used by R.H. Whittaker includes
(a) Body organisation
(b) Mode of nutrition
(c) Phylogenetic relationship
(d) Simple morphological characters

(1) (a), (b) and (d) only

(2) (a) and (b) only

(3) (a), (b) and (c) only

(4) All (a), (b), (c) and (d)

143. Sac fungi differs from imperfect fungi as the former

(1) Have septate and branched mycelium

(2) Produce conidia

(3) Have parasitic members

(4) Have sexual spores

144. Mark the **incorrect** statement w.r.t. binomial nomenclature.

(1) The genus and specific epithet collectively form binomial epithet along with author citation

(2) The name of discoverer is printed in italics

(3) Specific epithet starts with a small letter

(4) Biological names are generally taken from Latin language

145. Which of the following structures of bacteria help them to attach to the rocks?

(1) Pili

(2) Fimbriae

(3) Flagella

(4) Basal body

146. The cell wall resembles a soap box in

(1) Diatoms

(2) Dinoflagellates

(3) Slime moulds

(4) *Gonyaulax*

147. Select the **correct** statement

(1) Extrinsic growth is a defining feature of life forms

(2) Metabolism is a defining feature of all living organisms without exception

(3) Reproduction is a defining feature of all organisms

(4) Properties of tissues are present in the constituent cells

148. Chromatophores help in

(1) Respiration

(2) Locomotion

(3) Photosynthesis

(4) Excretion

149. The size of a typical eukaryotic cell is

(1) 1-2 µm

(2) About 3.0 µm

(3) 0.02 – 0.2 µm

(4) 10-20 µm

150. Family which includes genera *Felis* and *Panthera* is

(1) Muscidae

(2) Felidae

(3) Canidae

(4) Convolvulaceae

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ZOOLOGY

SECTION-A

151. Which of the cellular junctions is not found in epithelial tissue?

(1) Tight Junction

(2) Synaptic junction

(3) Gap junction

(4) Adhering junction

152. All of the following are the characteristic features of lining of alveoli, **except**

(1) A single thin layer of flattened cells with irregular boundaries

(2) Cells are compactly packed with little intercellular matrix

(3) The cells are ciliated

(4) The nuclei of cells are flat and often lie at the centre of cells

153. Bones are characterised by all, **except** that

(1) Its matrix is rich in calcium salts

(2) Its ground substance is rich in collagen fibres

(3) They are main tissue that provide structural frame to the body

(4) They have solid and pliable matrix

154. We are not able to contract intestinal muscles merely by thinking about it. Which of the following is not true for these muscle fibres?

(1) Fibres are fusiform in appearance

(2) Associated with visceral organs

(3) Absence of communication junctions

(4) Uninucleated condition

155. Which of the following is incorrect w.r.t. neural tissue?

(1) Neural tissue exerts the greatest control over the body's responsiveness to changing conditions

(2) Neuroglia are excitable cells

(3) Dendrites carry the nerve impulse towards the cell body

(4) Neuron is the structural and functional unit of neural system

156. The type of connective tissue that joins bone to bone is

(1) Dense irregular

(2) Dense regular

(3) Adipose tissue

(4) Specialized connective tissue

157. Brush bordered epithelium is found in

(1) Fallopian tube

(2) Small intestine

(3) Stomach

(4) Trachea

158. Choose the corret option to Complete the analogy.
Homopolymer : Cellulose : : Heteropolymer : _____

(1) Starch

(2) Chitin

(3) Inulin

(4) Insulin

159. Adenylic acid can be converted to adenosine by removing

(1) Ribose sugar

(2) 2'-deoxyribose sugar

(3) Adenine

(4) Phosphate group

160. In all of the given carbon compounds, heterocyclic rings can be found, **except**

(1) Cytosine

(2) Cysteine

(3) Cytidine

(4) Uridine

161. The EC number of an enzyme is 3.7.1.1. To which main class of enzymes does it belong?

(1) Hydrolases

(2) Lyases

(3) Ligases

(4) Transferases

162. The bond usually present in secondary structure of protein is

(1) Glycosidic bond

(2) Hydrogen bond

(3) Ester bond

(4) Ionic bond

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163. Bond formed by loss of a water molecule between two sugar molecules is called

(1) Peptide bond

(2) Glycosidic bond

(3) Phosphodiester bond

(4) Ester bond

164. Which of the following is **incorrect** for an exothermic reaction in presence of a biological catalyst?

(1) Enzyme decreases activation energy of reaction

(2) Potential energy of product is less than that of substrate

(3) Enzyme increases velocity of reaction

(4) Enzyme substrate complex does not pass through transition state and product is formed immediately.

165. Which of the following is acidic amino acid?

(1) Glycine

(2) Serine

(3) Cysteine

(4) Glutamic acid

166. Which of the following acts as a prosthetic group for peroxidase?

(1) Haem

(2) Zinc

(3) Niacin

(4) Copper

167. A conjugated lipid which is made up of glycerol with a phosphate group and two fatty acid is found in

(1) Cell membrane

(2) Silk

(3) Collagen

(4) Exoskeleton of arthropods

168. The process of inspiration is associated with all of the following, **except**

(1) Contraction of diaphragm

(2) Contraction of internal inter-costal muscles

(3) Increase in thoracic volume

(4) Lifting up of ribs and sternum

169. Total Lung Capacity (TLC) is **not** denoted by

(1) TLC = VC + RV

(2) TLC = IC + ERV

(3) TLC = IC + FRC

(4) TLC = TV + IRV + ERV + RV

170. The gaseous product of cellular respiration is carried by haemoglobin as

(1) Carbamino-haemoglobin

(2) Oxyhaemoglobin

(3) Methaemoglobin

(4) Carboxyhaemoglobin

171. In earthworms, respiration takes place through

(1) Moist skin

(2) Gills

(3) Lungs

(4) Tracheal system

172. Pleural fluid present between the pleura is responsible for

(1) Increasing friction on the lung surface

(2) Maintaining friction in the lung alveoli

(3) Decreasing friction on the lung surface

(4) Restoring friction between lungs

173. A person named 'XYZ' is blowing air into a balloon and forcibly exhales to fill up it. The volume of air which will not be able to leave his lungs and enter the balloon is

(1) Expiratory Reserve Volume

(2) Tidal Volume

(3) Residual Volume

(4) Inspiratory Reserve Volume

174. Below given are the features of 'X' involved in the regulation of respiration.
(a) Neural signals from this centre can reduce the duration of inspiration.
(b) It can moderate the functions of respiratory rhythm centre.
(c) It is present in the pons region of the brain.
Select the option that correctly identifies 'X'.

(1) Pneumotaxic centre

(2) Respiratory rhythm centre

(3) Chemosensitive area

(4) Appetite centre

175. The pO_2 in systemic veins of an adult healthy man is

(1) More than the pCO_2 of systemic arteries

(2) Less than the pCO_2 in alveolus

(3) Equal to the pO_2 in pulmonary vein

(4) Less than the pO_2 of systemic arteries

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176. **Assertion:** At the level of lungs, O_2 reaches blood from alveoli by crossing two cellular layers.
Reason: CO_2 from blood reaches into alveoli by crossing three layers *i.e.* endothelium of blood capillaries, basement membrane and alveolar membrane.
In the light of above statements, choose the **correct** option.

(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

177. Match column I with column II.

	Column I		Column II
a.	Tunica externa	(i)	Smooth muscles
b.	Tunica media	(ii)	Connective tissue
c.	Tunica interna	(iii)	Flattened squamous epithelial cells

Select the **correct** option.

(1) a(i), b(ii), c(iii)

(2) a(ii), b(i), c(iii)

(3) a(iii), b(ii), c(i)

(4) a(i), b(iii), c(ii)

178. **Assertion (A):** Wall of right atrium is thickest among the heart chambers.
Reason (R): Left ventricle pumps oxygenated blood to the body.
In the light of above statements, choose the correct answer from the following given options.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

179. 55% of the human blood is constituted by

(1) Formed elements

(2) Blood corpuscles

(3) Plasma

(4) RBCs

180. Thin cord-like structures called chordae tendineae are present in

(1) Atria

(2) Inter-atrial septum

(3) Ventricles

(4) Inter-ventricular septum

181. A person having only anti-B antibodies in his plasma, can donate blood to people with blood groups

(1) A and AB

(2) Only B

(3) O and B

(4) O and A

182. With the help of ECG, we can determine the rate of heart beat of an individual by counting the number of _____ that occur in a given time period.
Select the option to fill the blank **correctly.**

(1) P-waves

(2) T-waves

(3) QRS complexes

(4) P-Q intervals

183. Incomplete double circulation is exhibited by

(1) Human

(2) Frog

(3) Peacock

(4) Shark

184. In animals, the circulatory pattern 'P' is considered to be more advantageous than the circulatory pattern 'Q'. The flow of fluid in former can be precisely regulated. 'P' is usually found in

(1) Arthropods

(2) Fishes

(3) Most of molluscs

(4) Echinoderms

185. Narrowing of lumen of arteries caused due to deposition of calcium, fat, cholesterol and fibrous tissue that affects blood supply to the myocardium of heart is called

(1) Angina

(2) Atherosclerosis

(3) Heart failure

(4) Cardiac arrest

SECTION-B

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186. Which of the following tissues faces either a body fluid or the outside environment?

(1) Muscular tissue

(2) Nervous tissue

(3) Epithelial tissue

(4) Connective tissue

187. The tissue that forms a shock-absorbing cushion around the eyeballs and kidneys is

(1) Epithelial tissue

(2) Adipose tissue

(3) Areolar tissue

(4) Dense connective tissue

188. Which of the following is correct w.r.t the properties of areolar and adipose tissues?

(1) Both are examples of dense connective tissue

(2) Both are present beneath the skin

(3) Both of them do not store fats

(4) Both of them show compact packing of fibres and fibroblasts

189. **Assertion :** All the products of an enzyme catalysed reaction are released by the enzyme.
Reason : The free enzyme is ready to bind to another molecule of the substrate.
In the light of above statements, choose the **correct** option.

(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

190. Biomolecule that is **not** obtained in acid soluble fraction is

(1) Glucose

(2) Alanine

(3) Cholesterol

(4) Ribose

191. Select the **incorrect** statement w.r.t enzymes.

(1) Enzymes require optimum temperature for maximum activity

(2) Enzymes are denatured at low temperature

(3) Bell-shaped curve is formed when enzyme activity at y-axis is plotted against temperature at x-axis

(4) Enzymes catalyse many irreversible reactions

192. Read the following statements.
Statement I: Glycogen is a polysaccharide composed of glucose as its monomeric units.
Statement II: In the polysaccharide chain of glycogen, the right end is non-reducing end and left end is reducing end.
Choose the **correct** option.

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement II is correct but statement I is incorrect

193. The factor that shifts oxygen–haemoglobin dissociation curve to the left side is

(1) High partial pressure of oxygen

(2) High partial pressure of carbon dioxide

(3) High body temperature

(4) High H^+ concentration

194. Aortic and carotid chemoreceptors detect changes in

(1) CO_2 and H^+ concentration

(2) Body temperature

(3) Partial pressure of CO_2 only

(4) O_2 concentration

195. Read the given statements in context of humans.
(a) The lungs are situated in the thoracic chamber which is an anatomically air tight chamber.
(b) The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the lung cavity.

(1) Only statement (a) is correct

(2) Only statement (b) is correct

(3) Both statements are correct

(4) Both statements are incorrect

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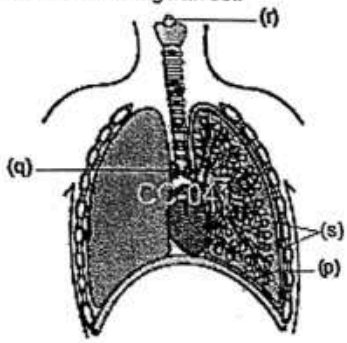
196. In the given diagrammatic view of human respiratory system, which structures can be related to the given description or function?

(A) Formed by division of windpipe and enters the lungs

(B) Pleural membranes, one of them is in close contact with the thoracic lining

(C) Epiglottis prevents entry of food into sound box during swallowing

(D) Bronchioles give rise to irregular walled and vascularised bags alveoli



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Select the **correct** option.

(1) p – (D), q – (A), r – (C), s – (B)

(2) p – (B), q – (D), r – (C), s – (A)

(3) p – (A), q – (D), r – (B), s – (C)

(4) p – (B), q – (C), r – (A), s – (D)

197. Number of double circulations completed by the human heart per minute under resting conditions is

(1) 36

(2) 72

(3) 144

(4) 18

198. Neural signals through the sympathetic nerve fibres can increase all of the following, **except**

(1) Heart rate

(2) Duration of cardiac cycle

(3) Cardiac output

(4) Speed of conduction of action potential

199. Which among the following set of WBCs are phagocytic in nature?

(1) Monocytes and Basophils

(2) Neutrophils and Monocytes

(3) Lymphocytes and Eosinophils

(4) Basophils and Eosinophils

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200. Lymph can be best described as

(1) Plasma without clotting factors

(2) Blood without RBCs, WBCs and platelets

(3) Blood minus RBCs and proteins

(4) Lymph is blood without larger plasma proteins and most of the formed elements

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